SHAMINDRA SHROTRIYA

Principal Data Scientist / AI Solution Architect / Applied AI Researcher

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I'm a hands-on principal data scientist with 13+ years spanning FMCG retail (Walmart, Quantium/Woolworths), consumer tech (freelancer.com), and actuarial insurance (PwC). I hold a Statistics PhD from Carnegie Mellon University (CMU). At Walmart, the world's largest retailer, I conceptualised modelling frameworks and carried them into production on Google Cloud. Flagship work includes *ShelfEdge 3.0*, a CEO-priority forecasting and allocation system I took from a 3-store pilot to 2,700 stores, saving ~5M labour hours a year. It also includes *Harmonization*, a patent-filed forecast alignment method I initiated, lifting accuracy by 4.7% (MAPE). At freelancer.com I helped build the A/B-testing framework and co-developed the causal methodology underpinning its evaluation, and at Quantium I built the behavioural segmentation still used in targeting campaigns for Woolworths customers, scored across ~7M shoppers at the time. I thrive on large-scale production challenges with no existing playbook, building solutions from first principles across methods from Bayesian time series to agentic GenAI. Above all I love the collaborative side of the work: leading and mentoring teams of data scientists and AI engineers, and presenting outcomes to director, VP, and senior-executive stakeholders.

EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
PhD in Statistics — Aug 2017 – Dec 2022
- **Thesis**: *On Some Problems in Nonparametric and Location-Scale Estimation.*
- **Nominee:** Umesh K. Gavaskar Best Thesis Award

Carnegie Mellon University — **Pittsburgh, PA**
M.S. in Statistics — Aug 2017 – Aug 2019
- Teaching Assistant of the Year.
- 3.97 / 4.0 GPA.

University of California, Berkeley — **Berkeley, CA**
M.A. in Statistics — Aug 2015 – Dec 2016
- Elizabeth Scott Memorial Award (Research).
- Outstanding Graduate Student Instructor (GSI) Award.
- 3.90 / 4.0 GPA.

University of New South Wales — **Sydney, NSW**
BCom in Actuarial Studies and Finance — Mar 2003 – Jan 2007
- UNSW Co-op scholar in Actuarial Studies.
- Graduated with Distinction.

SKILLS, LANGUAGES, INTERESTS

- **Programming**: Python, R, Julia, SQL, Bash/Shell (Unix), Make.
- **Agentic AI & LLM Tooling**: Claude Code, LangChain, Anthropic API, Vertex AI.
- **ML Engineering & Cloud Infrastructure**: PyTorch, JAX, scikit-learn, Google Cloud Platform (GCP), Apache Airflow, Docker.
- **Data & Dashboarding**: NumPy, pandas, polars, BigQuery, AWS Redshift, DuckDB, Git, marimo, Shiny, Streamlit.

WORK EXPERIENCE

Walmart, Inc (Retail Intelligence) — **Bentonville, AR**
Principal AI Solution Architect (promoted) — Apr 2024 – Mar 2026

Hands-on technical lead across concurrent flagship projects, conceptualising modelling frameworks and building the initial prototypes, then owning the methods through to production at scale. Led and mentored teams of up to 12 data scientists and engineers from junior through Principal level, reporting outcomes to director, senior-director, and VP stakeholders.

- Built *ShelfEdge 3.0*, a CEO-priority supply-chain forecasting and backroom-allocation system, by reframing its hard bounded-regression allocation problem as discretised classification to hit a two-month deadline. Designed the forecasting ensemble of ARMA and Bayesian DLMs with time-series foundation models (TimeGPT, NeuralForecast), then scaled the system from a 3-store pilot to all 2,700 robotics-equipped US stores on Apache Airflow.
- *ShelfEdge* saves an estimated 5M labour hours a year by predicting the backroom-versus-shop-floor split that feeds Walmart's fastunloader robotics, and cut demand-forecast error by 4.2% (MAPE).
- Initiated and led *Harmonization*, a novel patent-filed method that uses dynamic time warping to align related forecasts. Deployed it across 4,700 US stores at near-zero marginal cost, lifting forecast accuracy by 4.7% (MAPE).

- Architected and productionised the omnichannel Customer Experience (CX) pipeline across 50+ retail categories on BigQuery and Apache Airflow, cutting end-to-end Python runtime from ~22 hours to ~2 hours (~10×). Ran a 10-store pilot with the CXE causal model showing that relaxing Walmart's delivery promise from one day to two significantly cut fulfilment cost with no adverse effect on revenue.
- Co-designed the Google Cloud architecture integrating Walmart's *Sidekick* GenAI assistant with an agent-based store simulator, contributing to a 50-store production pilot in which *Sidekick* cut store team-leads' shift-planning time from 90 to 30 minutes.

Principal Data Scientist Sep 2023 – Apr 2024

- Co-defined Walmart's Customer Experience (CX) score, an omnichannel north-star metric built on customer retention and average basket size as a continuous, large-scale alternative to survey-based NPS. Designed the CX Elasticity (CXE) causal-measurement approach, including its breakdown across in-store, online, pickup, and delivery channels.
- Led an industry-academic collaboration with Prof. Giovanni Petris (University of Arkansas) on Walmart-scale Bayesian Dynamic Linear Model forecasts. A five-store pilot lifted workforce-forecast accuracy and cut understaffing by 1.4%.

Carnegie Mellon University (Statistics & Data Science) **Pittsburgh, PA**

Co-Director (CMU-Optum Bridges) and Assistant Teaching Professor, SURE Program Feb 2023 – Aug 2023

- Co-taught daily statistics and machine-learning lectures to the 30 nationally selected students of CMU's eight-week Summer Undergraduate Research Experience (SURE), across its sports analytics (CMSACamp) and healthcare (CMU-Optum Bridges) streams, and designed a new data-engineering course covering Unix, SQL, and MLOps fundamentals.
- Led the 15-student Optum healthcare stream, supervising four TA-mentored research teams through to final papers and posters. Nearly half the students earned Optum internships the following year and Optum renewed the program for two further summers.

freelancer.com **Sydney, NSW**

Team Lead, Data Science Infrastructure Jun 2014 – Jul 2015

- Co-developed the causal methodology behind the company's A/B-testing framework and helped build it from scratch in Python on AWS Redshift, serving ~15M daily active users (DAUs), owning hypothesis-test design and success metrics. The framework saved ~$150K a year over a private-vendor alternative and its pilot campaign delivered a 3.5% conversion-rate uplift.
- Co-led the ETL pipeline build into AWS Redshift during post-IPO international scaling, growing the data warehouse from ~11M to ~15M DAUs across 35+ countries.

Quantium (Woolworths Ltd.) **Sydney, NSW**

Lead Data Scientist Jun 2012 – Jun 2014

- Initiated and led the *Behavioural Lifestage* classifier, a seven-class family-structure segmentation built from Woolworths shopping behaviour and scored across the full ~7M-customer base. It became the primary targeting segmentation for major marketing campaigns across Australia and the backbone for multivariate promotion testing by family group.
- Cut ground-truth labelling costs by ~$110K a year by ageing prior survey labels forward three years to re-derive them for free, and by embedding an opt-in survey question in an existing Christmas promotion mailing, acquiring rich labelled data at marginal cost.
- Co-designed the NAB Online Retail Sales Index (*NORSI*), Australia's leading online-sales indicator, and developed its attribution framework decomposing month-to-month index movements into brand- and category-level drivers for NAB's economists. The index remains in monthly publication 13 years on, tracking ~$65B in annual online retail spend, around 15% of total Australian retail trade.
- Designed Woolworths Life Insurance's data-driven marketing strategy, building GLM scoring models and running A/B tests to lift customer response rates.

United Nations, International Labour Organization (ILO) **Brisbane, QLD**

Microinsurance (Actuarial) Fellow Jun 2011 – May 2012

- Selected as an ILO Microinsurance Fellow on industry experience and academic merit, and authored a report on actuarial pricing practices for microinsurance organisations.
- Represented the ILO at the 7th International Microinsurance Conference in Rio de Janeiro, a global forum of 400-plus practitioners and regulators from over 50 countries.

PwC (PricewaterhouseCoopers) **Sydney, NSW**

Senior Actuarial Consultant Jul 2007 – May 2011

- Conducted quarterly actuarial reserving for NSW WorkCover, Australia's largest public-sector workers compensation scheme (~$16B AUD total liability).

RESEARCH AND PUBLICATIONS

Nonparametric Estimation in the Dynamic Bradley-Terry Model.
AISTATS (Heejong Bong, Wanshan Li*, Shamindra Shrotriya*, Alessandro Rinaldo)* Jun 2020

Supremum-norm Bounds of the MLE in the BTL Model under General Comparison Graphs.
UAI (Wanshan Li, Shamindra Shrotriya*, Alessandro Rinaldo)* Jun 2022

Predictive Inference of a Wildfire Risk Pipeline in the United States.
NeurIPS Workshop on Tackling Climate Change with AI (Niccolo Dalmasso, Shamindra Shrotriya*, Alex Reinhart)* Dec 2019

WORKING PAPERS (UNDER REVIEW)

Revisiting Le Cam's Equation - Exact Minimax Rates over Convex Density Classes.
ArXiv:*2210.11436 (Shamindra Shrotriya*, Matey Neykov)* Jun 2024

Adversarial Sign-Corrupted Isotonic Regression.
ArXiv:*2207.07075 (Shamindra Shrotriya*, Matey Neykov)* Dec 2023

maars: Tidy Inference under the 'Models as Approximations' Framework in R.
ArXiv:*2106.11188 (Riccardo Fogliato*, Shamindra Shrotriya*, Arun Kumar Kuchibhotla)* Jun 2023

OPEN-SOURCE SOFTWARE

maars (R implementation of 'Models as Approximations' Framework)
https://shamindras.github.io/maars (Riccardo Fogliato, Shamindra Shrotriya*, Arun Kumar Kuchibhotla)* Jun 2023

iterative Random Forests (iRF)
https://github.com/Yu-Group/iterative-Random-Forest (joint work with the Yu Group at UC Berkeley) Aug 2017

HONORS AND AWARDS (SELECTED)

Nominee - Umesh K. Gavaskar Best Thesis Award **Pittsburgh, PA**
Issued by *Carnegie Mellon University* Jun 2023

Posit Scholar **Pittsburgh, PA**
Issued by *Posit, Inc (formerly RStudio)* Jun 2021

NGC Wildfire Research Scholar **New York, NY**
Issued by *American Australian Association* Jun 2020

Teaching Assistant of the Year **Pittsburgh, PA**
Issued by *Carnegie Mellon University* Jun 2020

CMSAC Best Paper Award **Pittsburgh, PA**
Issued by *Carnegie Mellon Sports Analytics Conference (CMSAC)* Oct 2019

Outstanding Graduate Student Instructor (GSI) **Berkeley, CA**
Issued by *University of California, Berkeley* Mar 2017

Elizabeth Scott Memorial Award (Research) **Berkeley, CA**
Issued by *University of California, Berkeley* Oct 2016

Geisinger Collider Project — Best Paper, Overall Winner **Pittsburgh, PA**
Issued by *Geisinger Marko Data Science Lab* Apr 2016

PRESENTATIONS AND TALKS

maars: Tidy Inference under misspecified statistical models in R Spotlight – Jul 2021
useR! 2021: The R Conference

Predictive Inference of a Wildfire Risk Pipeline in the United States Spotlight – Dec 2019
NeurIPS 2019 Workshop on Tackling Climate Change with AI

Efficient Estimation of Distribution-free Dynamics in the Bradley-Terry Model Talk – Nov 2019
Carnegie Mellon Sports Analytics Conference (CMSAC)

TEACHING EXPERIENCE

STAT 36-350 (Statistical Computing) 2018–2020
Head Teaching Assistant, Carnegie Mellon University
Instructors: Prof. Peter Freeman (2019, 2020), Prof. Ryan Tibshirani (2018)

- Led a teaching team of `6-7` undergraduate TAs and `2-3` Masters graduate TAs across three consecutive ~`200`-student cohorts. Owned grading on Gradescope / Canvas.
- Developed `R` course materials covering `tidyverse`, `ggplot2`, `dplyr` and authored sample-solution notebooks.

STAT 36-700 (Theoretical Statistics) 2018

Head Teaching Assistant, Carnegie Mellon University

Instructor: Prof. Larry Wasserman

- Led 2 PhD TAs across the ~80-student foundational theoretical statistics course. Wrote HW solutions and contributed to HW / exam design.

STAT 133 (Concepts in Computing with Data) 2016

Head Graduate Student Instructor (GSI), University of California, Berkeley

Instructor: Dr. Gaston Sanchez

- Led the TA team and grading for ~300 students in Berkeley Statistics' foundational statistical computing course.
- Held weekly R tutorial sessions.

SERVICE

Program Committee 2020

Climate-Change ML Workshop @ NeurIPS 2020

- Peer-reviewed workshop submissions.

Program Committee 2020

Climate-Change ML Workshop @ ICLR 2020

- Peer-reviewed workshop submissions.